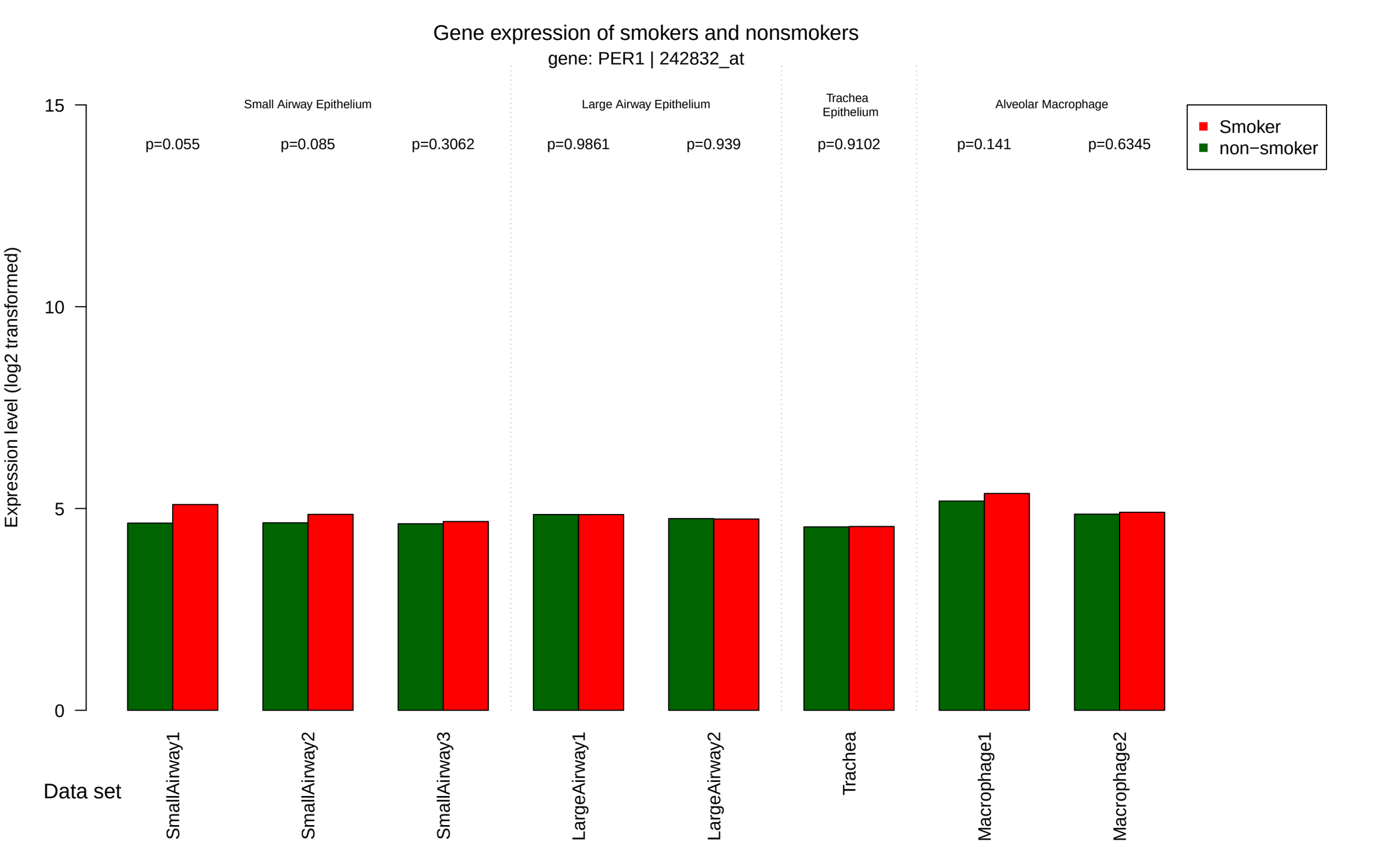
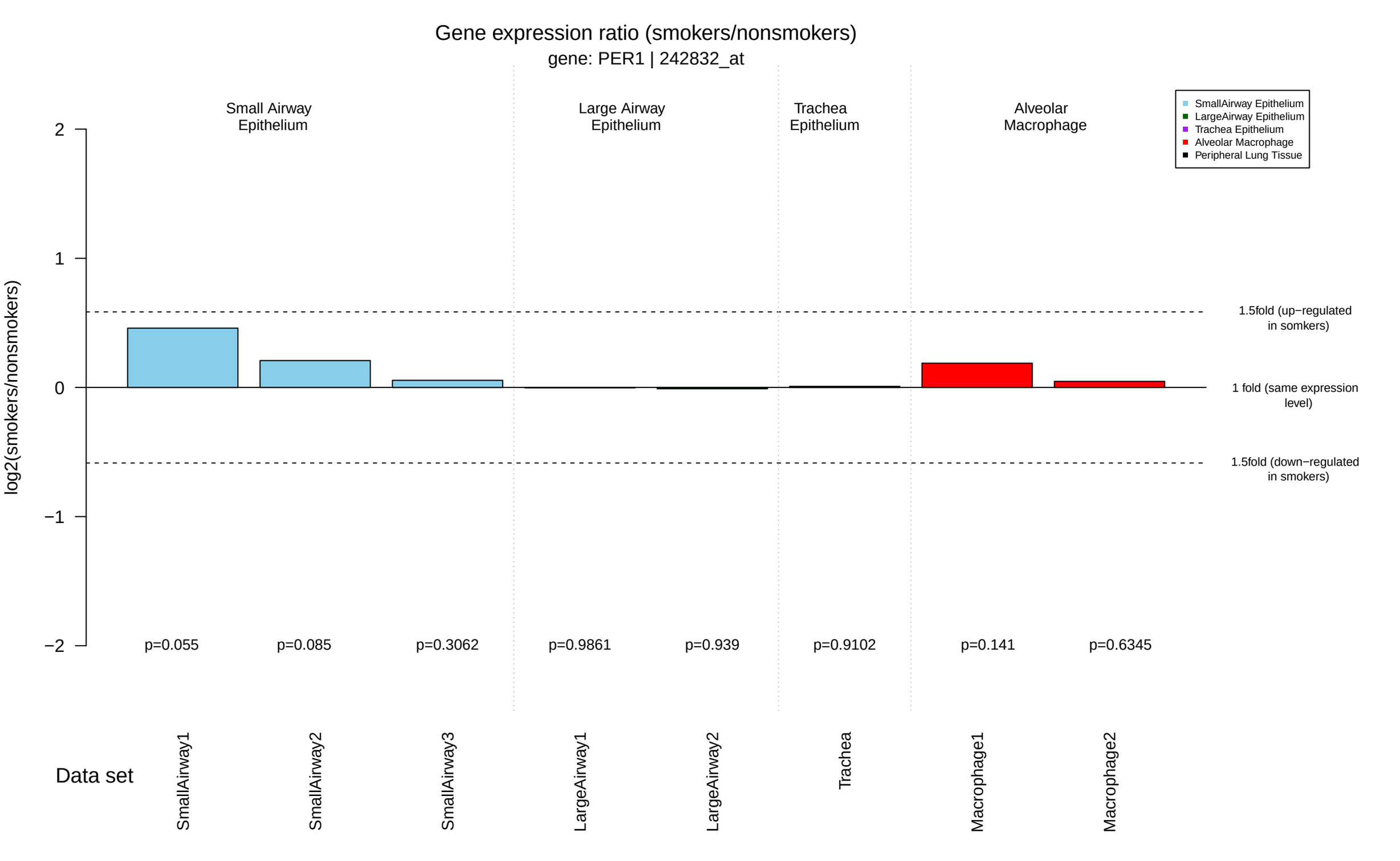
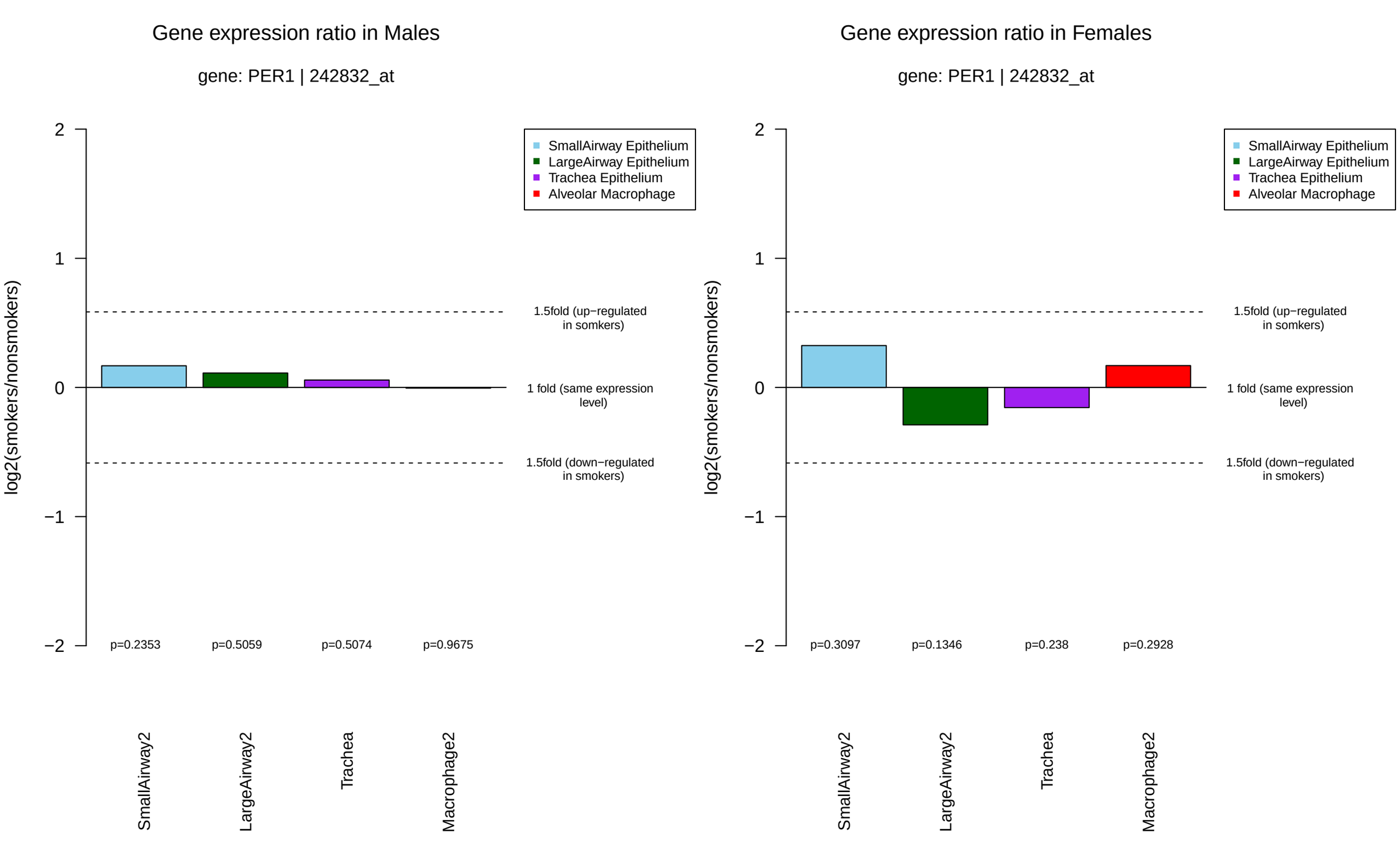
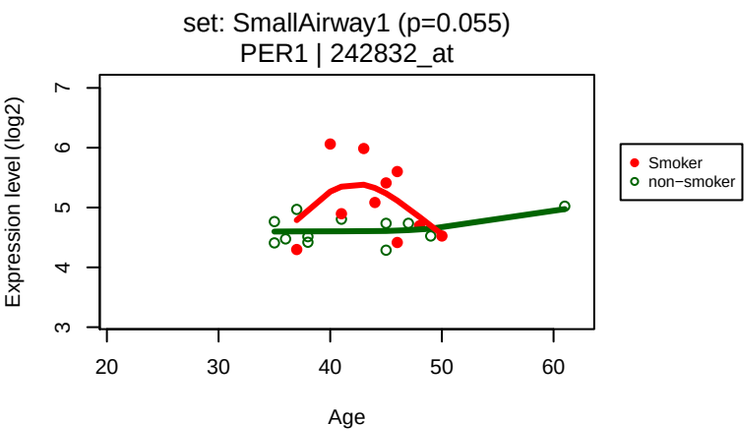
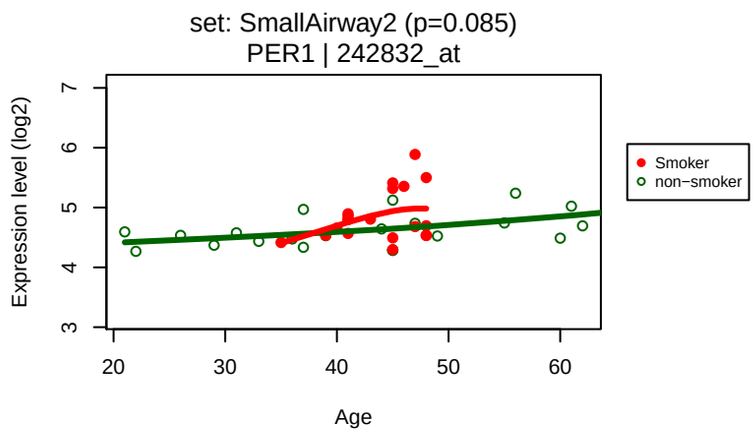
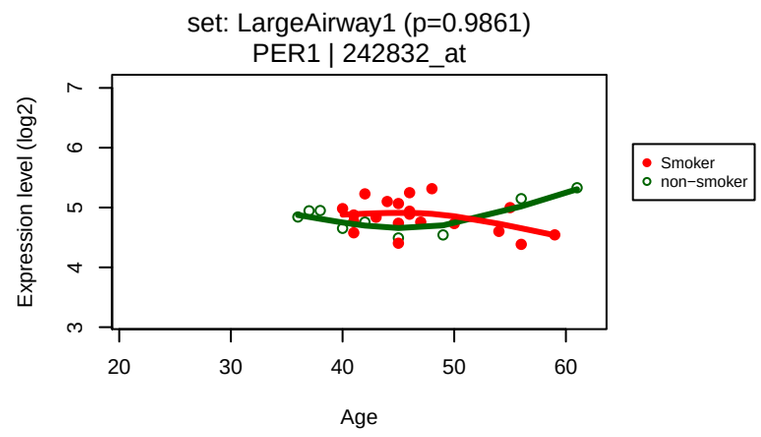

Gene expression of smokers and nonsmokers
gene: PER1 | 242832_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.055
p=0.085
p=0.3062
p=0.9861
p=0.939
p=0.9102
p=0.141
p=0.6345
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

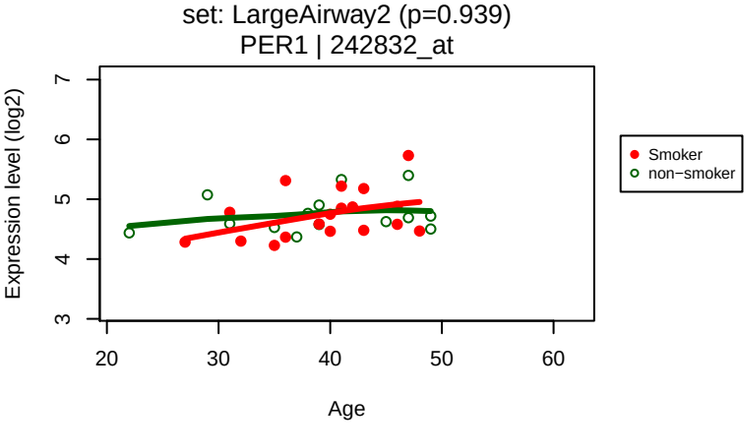

Gene expression ratio (smokers/nonsmokers)
gene: PER1 | 242832_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.055
p=0.085
p=0.3062
p=0.9861
p=0.939
p=0.9102
p=0.141
p=0.6345
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

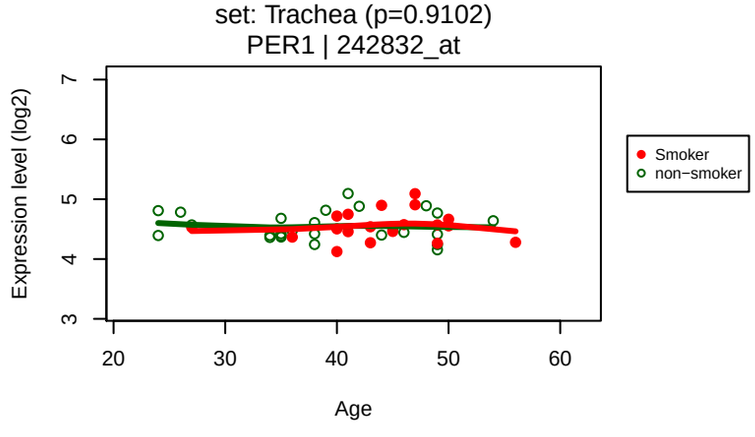

Gene expression ratio in Males
gene: PER1 | 242832_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2353
p=0.5059
p=0.5074
p=0.9675
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PER1 | 242832_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3097
p=0.1346
p=0.238
p=0.2928
SmallAirway2
LargeAirway2
Trachea
Macrophage2

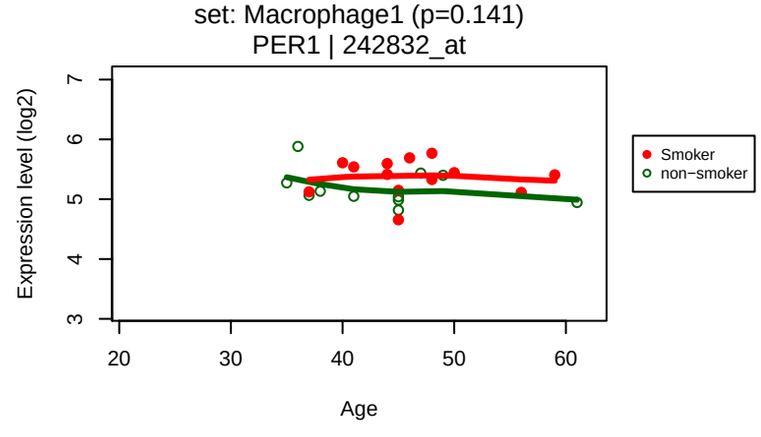

set: SmallAirway1 (p=0.055)
PER1 | 242832_at
Expression level (log2)
Age
Smoker
non−smoker

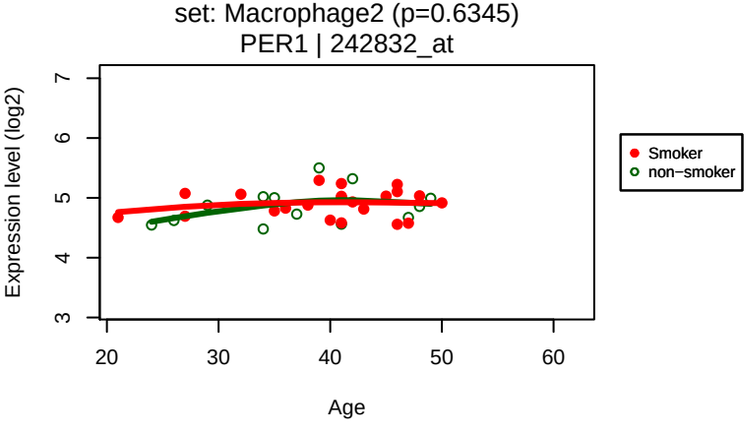

set: SmallAirway2 (p=0.085)
PER1 | 242832_at
Expression level (log2)
Age
Smoker
non−smoker

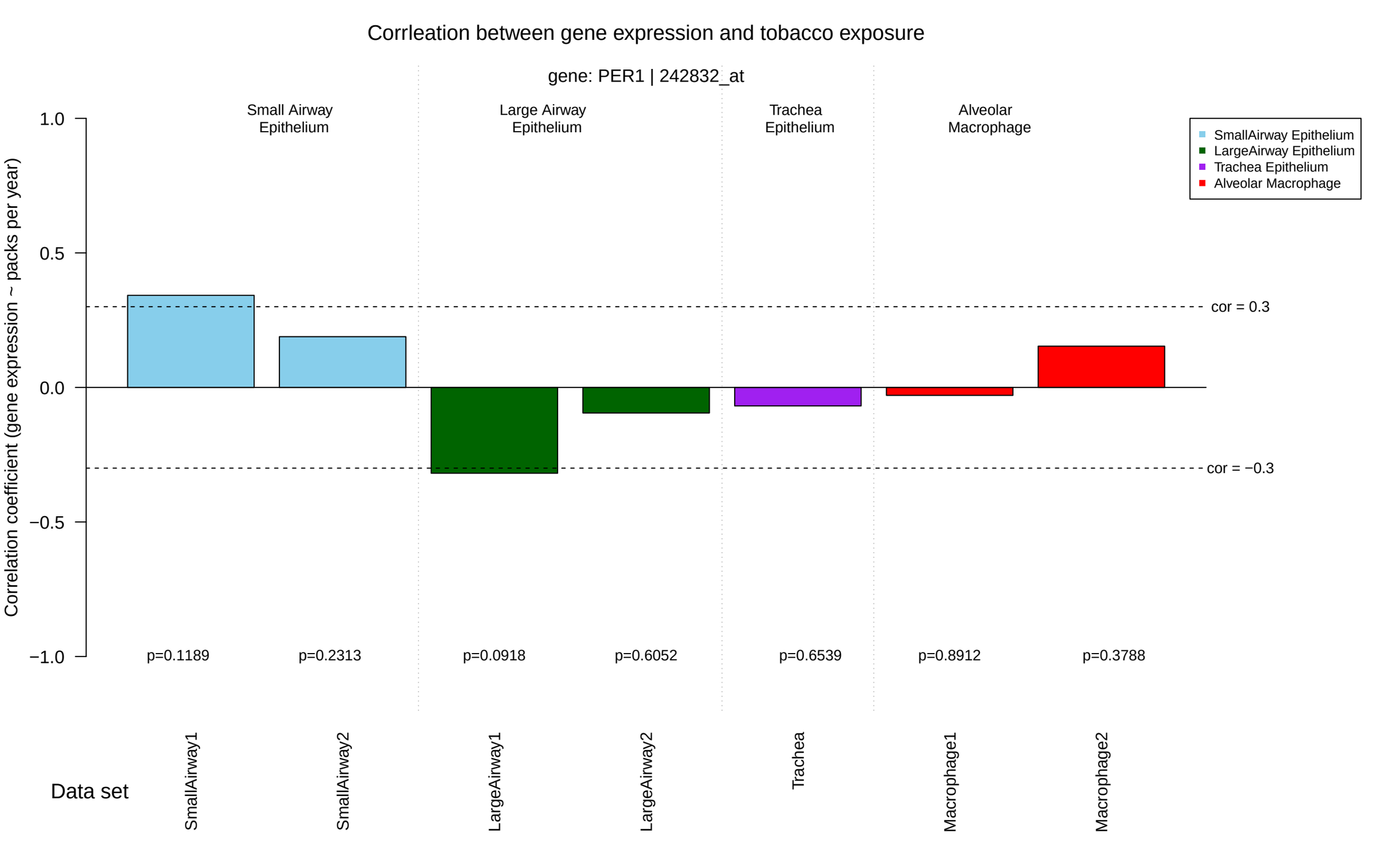

set: LargeAirway1 (p=0.9861)
PER1 | 242832_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.939)
PER1 | 242832_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9102)
PER1 | 242832_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.141)
PER1 | 242832_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6345)
PER1 | 242832_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PER1 | 242832_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1189
p=0.2313
p=0.0918
p=0.6052
p=0.6539
p=0.8912
p=0.3788
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2